



JKBOSE 2026

CLASS 11TH

LIMITS AND DERIVATIVES

MATHS

LECTURE-2

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TOPICS TO BE COVERED

1

Methods for Evaluation of Limits

2

Related Problems

$$\lim_{x \rightarrow 1} (x^2 + 1) = 2.$$

$$\lim_{x \rightarrow 5} \left(\frac{x^2 - 25}{x - 5} \right)$$

$$\lim_{x \rightarrow a} f(x) = l$$

$$f(x) \rightarrow l \text{ as } x \rightarrow a \left(x \rightarrow a^+ \text{ or } x \rightarrow a^- \right)$$

$$\lim_{x \rightarrow a^+} f(x) = \lim_{x \rightarrow a^-} f(x) \Rightarrow \lim_{x \rightarrow a} f(x) \text{ exists.}$$

$$\lim_{x \rightarrow a^+} f(x) \neq \lim_{x \rightarrow a^-} f(x) \Rightarrow \lim_{x \rightarrow a} f(x) \text{ DNE.}$$

6

Methods of Evaluating Limits

* Direct Substitution:

Replace x by a in the given function provided the replacement yields a finite number.

Example: Find the limit of the following functions at the indicated points:

(i) $f(x) = x^2 + 4$; at $x = 4$.

(ii) $f(x) = \frac{2x}{x^2 + 1}$, at $x = 1$.

(Sol) (i) Here $\lim_{x \rightarrow 4} f(x) = \lim_{x \rightarrow 4} (x^2 + 4) = 20$.

(ii) Here $\lim_{x \rightarrow 1} f(x) = \lim_{x \rightarrow 1} \frac{2x}{x^2 + 1} = \frac{2 \times 1}{1^2 + 1} = \frac{2}{2} = 1$.

* Factorisation:

Example: Find the limit of the function $f(x) = \frac{x^2 - 25}{x - 5}$, $x \neq 5$ at $x = 5$.

(Sol) Here $\lim_{x \rightarrow 5} f(x) = \lim_{x \rightarrow 5} \left(\frac{x^2 - 25}{x - 5} \right)$ | $\frac{0}{0}$ form.

$\Rightarrow \lim_{x \rightarrow 5} f(x) = \lim_{x \rightarrow 5} \left(\frac{x^2 - 5^2}{x - 5} \right) = \lim_{x \rightarrow 5} \frac{(x + 5)(x - 5)}{(x - 5)}$

$\Rightarrow \lim_{x \rightarrow 5} f(x) = \lim_{x \rightarrow 5} (x + 5) = 5 + 5 = 10$.

$\Rightarrow \boxed{\lim_{x \rightarrow 5} f(x) = 10}$ Hence the result.

EXAMPLE

Find the Limit of the following functions at the indicated points

$$(i) \quad f(x) = \frac{x^2 - 1}{x - 1}, \quad x \neq 1 \quad (x = 1)$$

$$(ii) \quad f(x) = \frac{x^2 - 5x + 6}{x - 3}, \quad x \neq 3 \quad (x = 3)$$

(Sol) (i) Here $\lim_{x \rightarrow 1} f(x) = \lim_{x \rightarrow 1} \left(\frac{x^2 - 1}{x - 1} \right) \left| \frac{0}{0} \text{ form} \right.$

$$\Rightarrow \lim_{x \rightarrow 1} f(x) = \lim_{x \rightarrow 1} \frac{(x+1)\cancel{(x-1)}}{\cancel{(x-1)}} = \lim_{x \rightarrow 1} (x+1)$$

$$\Rightarrow \boxed{\lim_{x \rightarrow 1} f(x) = 2}$$

(ii) (Sol) Here $\lim_{x \rightarrow 3} f(x) = \lim_{x \rightarrow 3} \left(\frac{x^2 - 5x + 6}{x - 3} \right) \left| \frac{0}{0} \text{ form} \right.$

$$\Rightarrow \lim_{x \rightarrow 3} f(x) = \lim_{x \rightarrow 3} \frac{(x-2)\cancel{(x-3)}}{\cancel{(x-3)}}$$

$$\Rightarrow \lim_{x \rightarrow 3} f(x) = \lim_{x \rightarrow 3} (x-2) = 3-2$$

$$\Rightarrow \boxed{\lim_{x \rightarrow 3} f(x) = 1} \quad \text{Hence the result.}$$

Example find the $\lim_{x \rightarrow \infty} \left(\frac{x^2 + 9x + 1}{x^2 + 2} \right)$.

(Sol) we have $\lim_{x \rightarrow \infty} \frac{x^2 + 9x + 1}{x^2 + 2}$ $\left| \frac{\infty}{\infty} \right.$ form

$$= \lim_{x \rightarrow \infty} \frac{\cancel{x^2} \left(1 + \frac{9}{x} + \frac{1}{x^2} \right)}{\cancel{x^2} \left(1 + \frac{2}{x^2} \right)} = \frac{(1 + 0 + 0)}{(1 + 0)}$$

$$\Rightarrow \lim_{x \rightarrow \infty} \left(\frac{x^2 + 9x + 1}{x^2 + 2} \right) = 1.$$

★ Rationalisation

Example Find the limit of the function $f(x) = \frac{\sqrt{x+2} - \sqrt{2}}{x}$

At $x = 0$.

(Sol) Here $\lim_{x \rightarrow 0} \left(\frac{\sqrt{x+2} - \sqrt{2}}{x} \right)$ $\left| \frac{0}{0} \right.$ form.

$$= \lim_{x \rightarrow 0} \frac{(\sqrt{x+2} - \sqrt{2})(\sqrt{x+2} + \sqrt{2})}{x(\sqrt{x+2} + \sqrt{2})} = \lim_{x \rightarrow 0} \frac{\cancel{x+2} - \cancel{2}}{x(\sqrt{x+2} + \sqrt{2})}$$

$$\Rightarrow \lim_{x \rightarrow 0} \left(\frac{\sqrt{x+2} - \sqrt{2}}{x} \right) = \lim_{x \rightarrow 0} \frac{\cancel{x+2} - \cancel{2}}{x(\sqrt{x+2} + \sqrt{2})} = \lim_{x \rightarrow 0} \frac{\cancel{x}}{\cancel{x}(\sqrt{x+2} + \sqrt{2})}$$

$$\Rightarrow \lim_{x \rightarrow 0} \left(\frac{\sqrt{x+2} - \sqrt{2}}{x} \right) = \frac{1}{\sqrt{2} + \sqrt{2}} = \frac{1}{2\sqrt{2}} \text{ Hence the result.}$$

EXAMPLE

Evaluate $\lim_{x \rightarrow 2} \left[\frac{1}{x-2} - \frac{2(2x-3)}{x^3-3x^2+2x} \right]$

(Sol) let $y = \lim_{x \rightarrow 2} \left(\frac{1}{x-2} - \frac{2(2x-3)}{x^3-3x^2+2x} \right)$ | $\infty - \infty$ form

$$\Rightarrow y = \lim_{x \rightarrow 2} \left(\frac{1}{x-2} - \frac{2(2x-3)}{x(x^2-3x+2)} \right)$$

$$\Rightarrow y = \lim_{x \rightarrow 2} \left(\frac{1}{x-2} - \frac{2(2x-3)}{x(x-1)(x-2)} \right)$$

$$\Rightarrow y = \lim_{x \rightarrow 2} \left\{ \frac{1}{(x-2)} \left(1 - \frac{2(2x-3)}{x(x-1)} \right) \right\}$$

$$\Rightarrow y = \lim_{x \rightarrow 2} \left\{ \frac{1}{(x-2)} \left(\frac{x(x-1) - 4x + 6}{x(x-1)} \right) \right\}$$

$$\Rightarrow y = \lim_{x \rightarrow 2} \frac{x^2 - x - 4x + 6}{x(x-1)(x-2)} = \lim_{x \rightarrow 2} \frac{x^2 - 5x + 6}{x(x-1)(x-2)} \quad \left| \frac{0}{0} \text{ form} \right|$$

$$\Rightarrow y = \lim_{x \rightarrow 2} \frac{(x-3)\cancel{(x-2)}}{x(x-1)\cancel{(x-2)}} = \lim_{x \rightarrow 2} \frac{(x-3)}{x(x-1)}$$

$$\Rightarrow y = \frac{2-3}{2(2-1)} = -\frac{1}{2}$$

$$\Rightarrow \boxed{y = -\frac{1}{2}}$$

EXAMPLE

Evaluate $\lim_{x \rightarrow a} \frac{\sqrt{a+2x} - \sqrt{3x}}{\sqrt{3a+x} - 2\sqrt{x}}$

(Sol) let $y = \lim_{x \rightarrow a} \frac{\sqrt{a+2x} - \sqrt{3x}}{\sqrt{3a+x} - 2\sqrt{x}} \quad \left| \frac{0}{0} \text{ form} \right|$

$$\Rightarrow y = \lim_{x \rightarrow a} \left(\frac{\sqrt{a+2x} - \sqrt{3x}}{\sqrt{3a+x} - 2\sqrt{x}} \right) \left(\frac{\sqrt{a+2x} + \sqrt{3x}}{\sqrt{a+2x} + \sqrt{3x}} \right)$$

$$\Rightarrow y = \lim_{x \rightarrow a} \frac{\cancel{\sqrt{a+2x}}^2 - \cancel{\sqrt{3x}}^2}{(\sqrt{a+2x} + \sqrt{3x})(\sqrt{3a+x} - 2\sqrt{x})}$$

$$\Rightarrow y = \lim_{x \rightarrow a} \frac{a+2x-3x}{(\sqrt{a+2x} + \sqrt{3x})(\sqrt{3a+x} - 2\sqrt{x})}$$

$$(2\sqrt{x})^2 = 4\sqrt{x}^2$$

$$\Rightarrow y = \lim_{x \rightarrow a} \frac{(a-x)}{(\sqrt{a+2x} + \sqrt{3x})(\sqrt{3a+x} - 2\sqrt{x})} \quad \left| \frac{0}{0} \text{ form} \right|$$

$$\Rightarrow y = \lim_{x \rightarrow a} \frac{(a-x)}{(\sqrt{a+2x} + \sqrt{3x})(\sqrt{3a+x} - 2\sqrt{x})} \left(\frac{\sqrt{3a+x} + 2\sqrt{x}}{\sqrt{3a+x} + 2\sqrt{x}} \right)$$

$$\Rightarrow y = \lim_{x \rightarrow a} \frac{(a-x)(\sqrt{3a+x} + 2\sqrt{x})}{(\sqrt{a+2x} + \sqrt{3x})(\sqrt{3a+x} - (2\sqrt{x})^2)}$$

$$\Rightarrow y = \lim_{x \rightarrow a} \frac{(a-x)(\sqrt{3a+x} + 2\sqrt{x})}{(\sqrt{a+2x} + \sqrt{3x})(3a+x-4x)}$$

$$\Rightarrow y = \lim_{x \rightarrow a} \frac{(a-x)(\sqrt{3a+x} + 2\sqrt{x})}{(\sqrt{a+2x} + \sqrt{3x})(3a-3x)}$$

$$\Rightarrow y = \lim_{x \rightarrow a} \frac{(a-x)(\sqrt{3a+x} + 2\sqrt{x})}{3(\sqrt{a+2x} + \sqrt{3x})(a-x)}$$

$$\Rightarrow y = \lim_{x \rightarrow a} \frac{\sqrt{3a+x} + 2\sqrt{x}}{3(\sqrt{a+2x} + \sqrt{3x})}$$

$$\Rightarrow y = \frac{\sqrt{3a+a} + 2\sqrt{a}}{3(\sqrt{a+2a} + \sqrt{3a})}$$

$$\Rightarrow y = \frac{\sqrt{4a} + 2\sqrt{a}}{3(\sqrt{3a} + \sqrt{3a})}$$

$$\Rightarrow y = \frac{2\sqrt{a} + 2\sqrt{a}}{3(2\sqrt{3a})}$$

$$\Rightarrow y = \frac{2\cancel{\sqrt{a}}}{3\cancel{2}\sqrt{3}\cancel{\sqrt{a}}} = \frac{2}{3\sqrt{3}}$$

$$\Rightarrow y = \frac{2}{3\sqrt{3}}$$

(Hence the result.)

EXAMPLE

$$\text{Define } f(x) = \begin{cases} 2x + 1 & \text{if } x < 1 \\ x^2 & \text{if } x \geq 1 \end{cases}$$

Find $\lim_{x \rightarrow 1^-} f(x)$, $\lim_{x \rightarrow 1^+} f(x)$, and $\lim_{x \rightarrow 1} f(x)$

(Sol) Here $\lim_{x \rightarrow 1^-} f(x) = \lim_{h \rightarrow 0} f(1-h)$

$$\Rightarrow \lim_{x \rightarrow 1^-} f(x) = \lim_{h \rightarrow 0} (2(1-h) + 1)$$

$$\Rightarrow \lim_{x \rightarrow 1^-} f(x) = \lim_{h \rightarrow 0} (2 - 2h + 1)$$

$$\Rightarrow \lim_{x \rightarrow 1^-} f(x) = \lim_{h \rightarrow 0} (3 - 2h) = 3 - 2 \times 0$$

$$\lim_{x \rightarrow a^-} f(x) = \lim_{h \rightarrow 0} f(a-h)$$
$$\lim_{x \rightarrow a^+} f(x) = \lim_{h \rightarrow 0} f(a+h)$$



$$\Rightarrow \lim_{x \rightarrow 1^-} f(x) = 3 \quad \text{--- (1)}$$

$$\text{Also, } \lim_{x \rightarrow 1^+} f(x) = \lim_{h \rightarrow 0} f(1+h)$$

$$\Rightarrow \lim_{x \rightarrow 1^+} f(x) = \lim_{h \rightarrow 0} (1+h)^2 = (1+0)^2$$

$$\Rightarrow \lim_{x \rightarrow 1^+} f(x) = 1 \quad \text{--- (2)}$$

Therefore from (1) & (2), we can conclude that $\lim_{x \rightarrow 1} f(x)$ does not exist.

Hence the result.

Methods of Evaluating Limits – Quick Reference

1. Direct Substitution	Function is defined and no indeterminate form	Plug in the value of x	$\lim_{x \rightarrow 3} (2x + 1) = 7$
2. Factoring	$\frac{0}{0}$ form, often in rational expressions	Factor and cancel	$\lim_{x \rightarrow 2} \frac{x^2 - 4}{x - 2}$ $= \lim_{x \rightarrow 2} \frac{(x - 2)(x + 2)}{x - 2} = 4$
3. Rationalizing	Radicals causing $\frac{0}{0}$	Multiply by conjugate	$\lim_{x \rightarrow 0} \frac{\sqrt{x + 1} - 1}{x}$
4. Standard Trig Limits	Involving $\sin x$, $\cos x$, etc.	Use known formulas	$\lim_{x \rightarrow 0} \frac{\sin x}{x} = 1$

QUESTION

What is the value of $\lim_{x \rightarrow 2} (x^2 + 3x - 4)$?

1

10 ✗

(Sol) Here it $(x^2 + 3x - 4)$
 $x \rightarrow 2$

2

8 ✗

$$\begin{aligned} &= 2^2 + 3 \times 2 - 4 \\ &= \cancel{4} + 6 - \cancel{4} \\ &= 6. \end{aligned}$$

3

6 ✓

4

4 ✗

QUESTION

Find the limit: $\lim_{x \rightarrow 1} \frac{x^2 - 1}{x - 1}$

1 $1 \times$

2 $0 \times$

3 $2 \checkmark$

4 Does not exist $\times$

$$\begin{aligned} & \lim_{x \rightarrow 1} \left(\frac{x^2 - 1}{x - 1} \right) \left| \frac{0}{0} \right. \\ &= \lim_{x \rightarrow 1} \frac{(x+1)\cancel{(x-1)}}{\cancel{(x-1)}} \\ &= \lim_{x \rightarrow 1} (x+1) = 2. \end{aligned}$$

QUESTION

Find $\lim_{x \rightarrow \infty} \frac{3x^2 + 5x + 1}{x^2 + 2}$

1

1 ✗

2

2 ✗

3

3 ✓

4

∞

(Sol) $\lim_{x \rightarrow \infty} \left(\frac{3x^2 + 5x + 1}{x^2 + 2} \right) \left| \frac{\infty}{\infty} \text{ form} \right|$

$= \lim_{x \rightarrow \infty} \frac{\cancel{x^2} \left(3 + \frac{5}{x} + \frac{1}{x^2} \right)}{\cancel{x^2} \left(1 + \frac{2}{x^2} \right)} = \frac{3 + 0 + 0}{1 + 0}$

$\Rightarrow \lim_{x \rightarrow \infty} \left(\frac{3x^2 + 5x + 1}{x^2 + 2} \right) = 3$

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THANK

You